JNTUH USED 23-06-18 AM Set 2

R13

Code No: 118BJ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2018
EMBEDDED SYSTEMS
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A

(25 Marks)

1.a) What is the timer flag interrupt? [2]
b) What is the UML notation for an object? [3]
c) What is the use MOVC instruction? [2]
d) What is description for SET TR1 instruction? [3]
e) What is shared data? [2]
f) How can differentiate RTOSs? [3]
g) Write some hard RTOS names. [2]
h) What is Linker? [3]
i) What are the features of memory organization? [2]
j) Draw the basic class diagram for the elevator system. [3]

PART - B

(50 Marks)

2.a) Provide a concrete example of how bottom–up information from the software programming phase of design may be useful in refining the architectural design.
b) Explain each pin functionality of port 3 8051 controller. [5+5]

OR

3.a) Briefly describe the distinction between requirements and specification.
b) Explain about mode 2 operation of timers. [5+5]

4.a) Write a simple subroutine , call it, and jump back to the calling program after adjusting the stack pointer.
b) Write a ALP program for square the content of R5, put the results in R0 (high byte) and R1 (low byte). [5+5]

OR

5.a) Assume the crystal frequency is 10megahertz , write a program that will use timer 1 to interrupt the program after a delay of 2ms.
b) Write a ALP program for multiply the data in RAM location 22h by the data in RAM location 15h, put the result in RAM lacationsm19h(low byte) and 1Ah (high byte). [5+5]

6. Explain about Interrupt Routines in an RTOS Environment. [10]

OR

7. Write a program using timer 0 that will delay exactly 100000 ms±1us from the time the timer start until it is stopped. [10]

8.a) What are Hard Real-Time Scheduling Considerations in RTOS? Explain.
b) Explain basic concept of Getting Embedded Software into the Target System. [5+5]

OR

9. Explain in detail about the following:
a) Saving Memory and power
b) Linkers/Locators for embedded software. [5+5]

10. Explain in detail about SHARC architecture. [10]

OR

11. Explain Applications and security in Internet-Enabled Systems. [10]

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